Name _____ _____ Baggett/Fall2026/207/Quiz5

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) If f and g are differentiable, then $\dfrac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$.

True **False**

Why? **The Chain Rule. Note where each factor is evaluated: the outer derivative at the INNER function's value, $g(x)$, and then multiplied by $g'(x)$.**

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2. (**2** pts) To differentiate implicitly you must first solve the equation for y.

True **False**

Why? **The whole point is that you do not. Differentiate both sides as they stand, treating y as a function of x, then solve for y'.**

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3. (**4** pts) **Derive** $\dfrac{d}{dx}\left[\csc x\right]$ starting from $\csc x = \dfrac{1}{\sin x}$. Use the Quotient Rule and **show every step**. Quoting the formula earns nothing.

$\dfrac{d}{dx}\left[\csc x\right] = -\csc x \cot x$

$$\frac{d}{dx}\left[\frac{1}{\sin x}\right] = \frac{\sin x \cdot \frac{d}{dx}(1) - 1 \cdot \frac{d}{dx}(\sin x)}{\sin^2 x} \qquad \text{lo D hi minus hi D lo, over lo squared}$$

$$= \frac{\sin x \cdot 0 - 1 \cdot \cos x}{\sin^2 x} \qquad \text{the top is a constant, so one term dies}$$

$$= -\frac{\cos x}{\sin^2 x} \qquad \text{nothing cancels the minus — it stays}$$

$$= -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = -\csc x \cot x \qquad \text{split it to name it}$$

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4. (**2** pts) Differentiate $f(x) = \sqrt{x^2+9}$. You do not need to simplify.

$f'(x) = \dfrac{x}{\sqrt{x^2+9}}$

$$f'(x) = \tfrac{1}{2}\left(x^2+9\right)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+9}} \qquad \text{Chain Rule}$$

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5. (**3** pts) Differentiate $g(t) = e^{-2t}\cos(3t)$. You do not need to simplify.

$g'(t) = -2e^{-2t}\cos(3t) - 3e^{-2t}\sin(3t)$

$$\begin{aligned} g'(t) &= \frac{d}{dt}\left[e^{-2t}\right]\cos(3t) + e^{-2t}\,\frac{d}{dt}\left[\cos(3t)\right] && \text{Product Rule first} \\ &= \left(-2e^{-2t}\right)\cos(3t) + e^{-2t}\left(-3\sin(3t)\right) && \text{each factor is its own Chain Rule} \\ &= -e^{-2t}\left(2\cos(3t) + 3\sin(3t)\right) && \text{it factors} \end{aligned}$$

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6. (**4** pts) Find $\dfrac{dy}{dx}$ by **implicit differentiation**:

$$x^2 + xy + y^3 = 5$$

$\dfrac{dy}{dx} = -\dfrac{2x+y}{x+3y^2}$

$$\begin{aligned} 2x + \left(y + x\,\frac{dy}{dx}\right) + 3y^2\,\frac{dy}{dx} &= 0 && xy \text{ is a Product; } y^3 \text{ carries a } \tfrac{dy}{dx} \\ x\,\frac{dy}{dx} + 3y^2\,\frac{dy}{dx} &= -2x - y && \text{gather} \\ \frac{dy}{dx}\left(x + 3y^2\right) &= -2x - y && \text{factor} \\ \frac{dy}{dx} &= -\frac{2x+y}{x+3y^2} && \text{divide} \end{aligned}$$

/4

7. (**3** pts) Differentiate $y = x^2\sec x$.

$y' = 2x\sec x + x^2\sec x\tan x$

$$\begin{aligned} y' &= \frac{d}{dx}\left[x^2\right]\sec x + x^2\,\frac{d}{dx}\left[\sec x\right] && \text{Product Rule} \\ &= 2x\sec x + x^2\sec x\tan x \end{aligned}$$

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